

# Sachin Bala

Sachin.bala@wmich.edu | 269-779-9757 | github.com/Sachin1bala1

## Technical Skills

Python (Pandas, NumPy, scikit-learn), SQL (MySQL), R, JMP, Statistical Analysis, DOE, SPC, Process Capability Studies, Data Visualization (Grafana, Matplotlib), Machine Learning, Computer Vision (YOLO), Minitab, LabVIEW, SolidWorks, AutoCAD, Six Sigma, PFMEA, Lean Manufacturing, Microsoft Suite, Cognex, Dr Schenk vision system

## Education

**Western Michigan University (WMU)**

**Overall GPA: 3.69**

**Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics**

## Work Experiences

**Associate Process Development Engineer, Tesla Inc., Austin, TX**

**02/25-present**

- Commissioned two electrode lines producing electrode film at 75 meters per minute, achieving a 98% yield rate through systematic process optimization and rigorous qualification protocols, with progress tracked via real-time operational metrics.
- Applied Six Sigma and statistical methodologies (SPC, DOE, FMEA, Capability Studies, 8D, 5-Why) to analyze large-scale process datasets, identify root causes of yield excursions, and drive measurable improvements in manufacturing efficiency.
- Engineered data collection, visualization, and analysis pipelines using MySQL, Python, R, and JMP to generate predictive insights into tool and process performance, enabling data-driven operational and financial decisions across the production organization.

**Engineering Technician, Process Development Team, Tesla Inc., Austin, TX**

**8/23-02/25**

- Designed and deployed a real-time monitoring pipeline in Grafana to track tool vibration metrics across the production floor, enabling proactive anomaly detection; authored SOPs for sensor integration and dashboard usage.
- Conducted statistical capability studies and qualification analyses to validate tools against product specifications, identifying data-driven process improvements that increased production rate by 10% while sustaining quality targets.
- Performed systematic root cause analysis on downtime events using structured data investigation (fault logs, SPC charts, process historians), implementing corrective actions that reduced unplanned stoppages and improved operational uptime.

**Engineering Co-Op, Scherdel SST, Muskegon, MI**

**8/22-12/22**

- Analyzed EV component market data and manufacturing material trends, synthesizing quantitative findings to support product launch decisions and quality process design.

**Process Engineering Intern, Scherdel SST, Muskegon, MI**

**05/22-08/22**

- Built a predictive model using ML algorithms and statistical data analysis to detect early spring manufacturing failures, improving production efficiency and accuracy by 25%.
- Developed a Computer Vision defect detection system; validated model performance using Gauge R&R methodology and F1-score analysis to ensure reliability.

**Laser Tech Technician Internship, Technique Inc., Jackson, MI**

**09/21-04/22**

- Optimized laser cutting using CAD and machine troubleshooting; validated chassis parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures.

## Engineering Research and Project

**Senior Design Project - Two Tier Industrial Cart (Technique Inc.)**

**01/22-12/22**

- Designed a two-tier industrial cart in SolidWorks with scissor lift mechanism; validated with FEA analysis and numerical calculations.
- Applied DFMEA to identify and mitigate risk, establishing data-backed design controls and a quality control plan.

## Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)